Physics 827: Homework Set No. 3

Due date: Thursday, October 13, 2011, 3:00pm in PRB M2043 (Biao Huang's office)

Total point value of set: 60 points

Problem 1 (20 pts.): Show that $e^{\hat{L}+\hat{M}} = e^{\hat{L}}e^{\hat{M}}e^{-\frac{1}{2}[\hat{L},\hat{M}]}$ if $[\hat{L},[\hat{L},\hat{M}]] = 0 = [\hat{M},[\hat{L},\hat{M}]]$.
Hint: Consider $\hat{F}(\alpha) = e^{\alpha(\hat{L}+\hat{M})}$. Show that $\hat{L}\hat{F} = \hat{F}\hat{L} - \alpha[\hat{L},\hat{M}]$ (see problem 9 of HW#2). Use this to bring $\frac{d\hat{F}}{d\alpha}$ into a form that you can integrate to give, for $\alpha = 1$, the r.h.s. of the relation you want to prove. Fill in the rest.

Problem 2 (20 pts.): Show that

$$\text{(i) } \lim_{\epsilon\to 0^+} \delta_\epsilon(x) \equiv \lim_{\epsilon\to 0^+} \begin{Bmatrix} \frac{1}{\epsilon} & \text{for} & |x| < \frac{\epsilon}{2} \\ 0 & \text{for} & |x| > \frac{\epsilon}{2} \end{Bmatrix} = \delta(x) \text{ (5 pts.),}$$

$$\text{(ii) } \lim_{\epsilon\to 0^+} \delta_\epsilon(x) \equiv \lim_{\epsilon\to 0^+} \frac{1}{\pi\epsilon}\left(\frac{\sin(x/\epsilon)}{x/\epsilon}\right)^2 = \delta(x) \text{ (15 pts.).}$$

If in part (ii) you can prove the result without evaluating any integrals with MATHEMATICA or by looking them up in an integral table, you earn **5 bonus points**.

Problem 3 (20 pts.): For a function $f(x) = \langle x|f\rangle$, x real, with $\langle f|f\rangle = 1$, compute the following matrix elements of the position operator $\hat{X}$ and momentum operator $\hat{P} = \hbar\hat{K}$ in the eigenbases of $\hat{X}$ and $\hat{K}$, respectively:
$\langle k|\hat{X}^2|f\rangle$; $\langle x|\hat{X}^2|f\rangle$; $\langle k|\hat{P}^2|f\rangle$; $\langle x|\hat{P}^2|f\rangle$; $\langle k|\frac{1}{2}(\hat{X}\hat{P}+\hat{P}\hat{X})|f\rangle$; and $\langle x|\frac{1}{2}(\hat{X}\hat{P}+\hat{P}\hat{X})|f\rangle$.

Problem 1

Consider $\hat{F}(\alpha) = e^{\alpha(\hat{L}+\hat{M})}$

Compute $\hat{F}(\alpha)\,\hat{L}\,\hat{F}^{-1}(\alpha) = e^{\alpha(\hat{L}+\hat{M})}\,\hat{L}\,e^{-\alpha(\hat{L}+\hat{M})}$ = (use problem 8, HW2)

$$= \sum_{n=0}^{\infty} \frac{1}{n!}\left[\alpha(\hat{L}+\hat{M}), \hat{L}\right]_{(n)} = \left[\alpha(\hat{L}+\hat{M}), \hat{L}\right]_{(0)} + \left[\alpha(L+M), \hat{L}\right]_{(1)} + \ldots$$

$= \hat{L} + \alpha[\hat{M},\hat{L}] + 0$ (since $[\hat{M},\hat{L}]$ commutes with both $\hat{L}$ and $\hat{M}$, all higher order terms are zero)

$$\Rightarrow \hat{F}\hat{L} = \hat{L}\hat{F} + \alpha[\hat{M},\hat{L}]\hat{F} = \hat{L}\hat{F} - \alpha[\hat{L},\hat{M}]\hat{F} \qquad (*)$$

Now compute $\dfrac{d\hat{F}}{d\alpha} = \hat{F}(\hat{L}+\hat{M}) = \hat{F}\hat{L} + \hat{F}\hat{M} \stackrel{(*)}{=} \hat{L}\hat{F} + \hat{F}\hat{M} - \alpha[\hat{L},\hat{M}]\hat{F}$

This is the derivative of

$$\hat{F}(\alpha) = e^{\alpha\hat{L}} e^{\alpha\hat{M}} e^{-\frac{1}{2}\alpha^2[\hat{L},\hat{M}]}$$

as is easily verified:

$$\frac{dF}{d\alpha} = \left(\hat{L}e^{\alpha\hat{L}}\right)e^{\alpha\hat{M}}e^{-\frac{\alpha^2}{2}[\hat{L},\hat{M}]} + e^{\alpha\hat{L}}\left(e^{\alpha\hat{M}}\hat{M}\right)e^{-\frac{\alpha^2}{2}[\hat{L},\hat{M}]} + e^{\alpha\hat{L}}e^{\alpha\hat{M}}\left(-\alpha[\hat{L},\hat{M}]e^{-\frac{\alpha^2}{2}[\hat{L},\hat{M}]}\right)$$

(since $[\hat{L},\hat{M}]$ commutes with both $\hat{M}$ and $\hat{L}$)

$$= \hat{L}\left(e^{\alpha\hat{L}}e^{\alpha\hat{M}}e^{-\frac{\alpha^2}{2}[\hat{L},\hat{M}]}\right) + \left(e^{\alpha\hat{L}}e^{\alpha\hat{M}}e^{-\frac{\alpha^2}{2}[\hat{L},\hat{M}]}\right)\hat{M}$$

$$- \alpha[L,M]\left(e^{\alpha\hat{L}}e^{\alpha\hat{M}}e^{-\frac{\alpha^2}{2}[\hat{L},\hat{M}]}\right)$$

$$= \hat{L}\hat{F} + \hat{F}\hat{M} - \alpha[\hat{L},\hat{M}]\hat{F}.$$

Hence $\dfrac{d}{d\alpha} e^{\alpha(\hat{L}+\hat{M})} = \dfrac{d}{d\alpha}\left(e^{\alpha\hat{L}}e^{\alpha\hat{M}}e^{-\frac{\alpha^2}{2}[\hat{L},\hat{M}]}\right)$

and $e^{\alpha(\hat{L}+\hat{M})}\Big|_{\alpha=0} = e^{\alpha\hat{L}}e^{\alpha\hat{M}}e^{-\frac{\alpha^2}{2}[\hat{L},\hat{M}]}\Big|_{\alpha=0}$

Since both sides agree at $\alpha=0$ and their derivatives agree for all α, the two sides are equal: $\boxed{e^{\alpha(\hat{L}+\hat{M})} = e^{\alpha\hat{L}}e^{\alpha\hat{M}}e^{-\frac{\alpha^2}{2}[\hat{L},\hat{M}]}}$

The desired result follows by setting $\alpha=1$.

Problem 2

(i) $$\delta_\varepsilon(x) = \begin{cases} \frac{1}{\varepsilon} & \text{for } |x| < \frac{\varepsilon}{2} \\ 0 & \text{for } |x| > \varepsilon/2 \end{cases}$$

In the limit $\varepsilon \to 0$, this function is ∞ at $x=0$ and 0 at $x \neq 0$. Remains to show that

$$\lim_{\varepsilon\to 0} \int f(x)\,\delta_\varepsilon(x)\,dx \stackrel{?}{=} f(0)$$

Ok, let's see:

$$\lim_{\varepsilon\to 0} \int f(x)\,\delta_\varepsilon(x)\,dx = \lim_{\varepsilon\to 0} \int_{-\varepsilon/2}^{\varepsilon/2} \frac{1}{\varepsilon} f(x)\,dx \qquad \text{(near}$$

$$= \lim_{\varepsilon\to 0} \frac{1}{\varepsilon} \int_{-\varepsilon/2}^{\varepsilon/2} \left[f(0) + x f'(0) + \tfrac{1}{2}x^2 f''(0) \ldots\right] dx$$

$$= f(0) \lim_{\varepsilon\to 0} \frac{1}{\varepsilon} \int_{-\varepsilon/2}^{\varepsilon/2} dx + f'(0) \lim_{\varepsilon\to 0} \frac{1}{\varepsilon} \int_{-\varepsilon/2}^{\varepsilon/2} x\,dx + \ldots$$

$$= f(0) \lim_{\varepsilon\to 0} \frac{1}{\varepsilon}\cdot\varepsilon + f'(0) \lim_{\varepsilon\to 0} \frac{1}{\varepsilon}\frac{\varepsilon^2}{2} + f''(0) \lim_{\varepsilon\to 0} \frac{1}{\varepsilon}\frac{\varepsilon^3}{4} + \ldots$$

$$= f(0) \checkmark$$

(ii) $$\delta_\varepsilon(x) = \frac{1}{\pi\varepsilon} \frac{\sin^2(x/\varepsilon)}{(x/\varepsilon)^2}$$

- For $x \to 0$, $\frac{\sin(x)}{x} \to 1$, hence $\lim_{x\to 0} \delta_\varepsilon(x) = \frac{1}{\pi\varepsilon} \xrightarrow[\varepsilon\to 0]{} \infty$

$$\Rightarrow \lim_{\varepsilon\to 0} \delta_\varepsilon(0) = \infty \checkmark$$

- For $x \neq 0$, $\lim_{\varepsilon\to 0} \delta_\varepsilon(x) = \lim_{\varepsilon\to 0} \frac{\sin^2(x/\varepsilon)}{\pi x^2/\varepsilon} = \lim_{\varepsilon\to 0} \frac{\varepsilon}{\pi} \frac{\sin^2(x/\varepsilon)}{x^2}$

$$\leq \lim_{\varepsilon\to 0} \frac{\varepsilon}{\pi}\frac{1}{x^2} = 0$$

$$\Rightarrow \lim_{\varepsilon\to 0} \delta_\varepsilon(x) = 0 \text{ for } x \neq 0 \checkmark$$

Remains to show that

$$\lim_{\varepsilon\to 0}\frac{1}{\pi\varepsilon}\int f(x)\,\frac{\sin(x/\varepsilon)}{(x/\varepsilon)^2}\,dx \overset{?}{=} f(0)$$

Ok, let's see:

$$\lim_{\varepsilon\to 0}\frac{1}{\pi\varepsilon}\int_{-\infty}^{\infty} f(x)\,\frac{\sin(x/\varepsilon)}{(x/\varepsilon)^2}\,dx \overset{t=\frac{x}{\varepsilon}}{=} \lim_{\varepsilon\to 0}\frac{1}{\pi}\int_{-\infty}^{\infty} dt\, f(\varepsilon t)\,\frac{\sin^2(t)}{t^2}$$

$$= f(0)\,\underbrace{\frac{1}{\pi}\int_{-\infty}^{\infty}\frac{\sin^2 t}{t^2}\,dt}_{1} = f(0)$$

The last step you can do by (a) using MATHEMATICA
(b) looking the integral up in Gradsteyn/Ryzhik or similar
(c) contour integration.

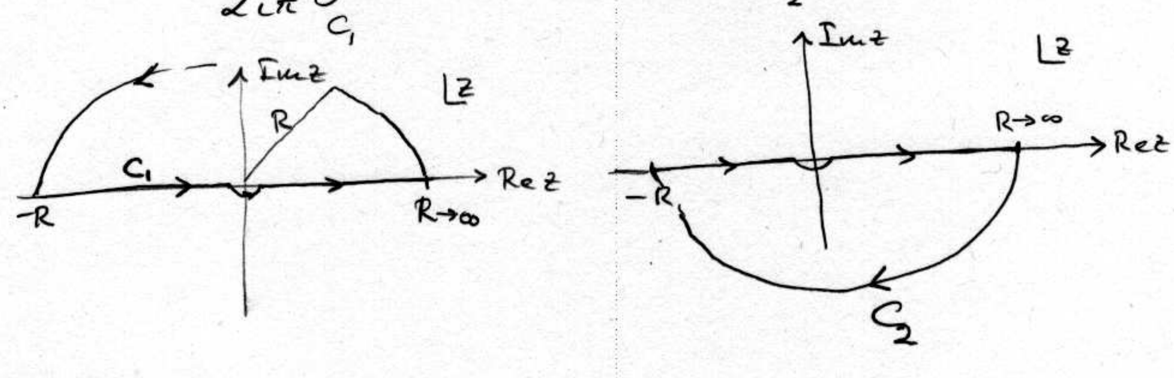
Let's illustrate path (c):

$$\frac{1}{\pi}\int_{-\infty}^{\infty} dt\,\frac{\sin^2 t}{t^2} \underset{\uparrow\ \text{by parts}}{=} \underbrace{-\frac{1}{\pi t}\sin^2 t\,\Big|_{\infty}^{\infty}}_{0} + \frac{2}{\pi}\int_{-\infty}^{\infty} dt\,\frac{\sin t\cos t}{t}$$

$$= \frac{1}{\pi}\int_{-\infty}^{\infty} dz\,\frac{\sin z}{z} \quad \text{where } z = 2t \quad (2\sin t\cos t = \sin 2t)$$

$$= \frac{1}{2i\pi}\left(\int_{-\infty}^{\infty}\frac{dz}{z}e^{iz} - \int_{-\infty}^{\infty}\frac{dz}{z}e^{-iz}\right)$$

$$= \frac{1}{2i\pi}\oint_{C_1}\frac{dz}{z}e^{iz} - \frac{1}{2i\pi}\oint_{C_2}\frac{dz}{z}e^{-iz}$$



In this step we have written $\frac{1}{z} = \lim_{\delta \to 0^+} \frac{1}{z - i\delta}$

(i.e. shifted the pole infinitesimally above the real axis, letting it approach the real axis at the end — you get the same result if you shift it infinitesimally below the real axis, $\frac{1}{z} = \lim_{\delta \to 0^+} \frac{1}{z + i\delta}$)

and added to the integral $\int_{-\infty}^{\infty} dz$ along the real axis (the integral we want to evaluate) a zero in the form of an integral over a semicircle of radius R along which the integrand vanishes in the limit $R \to \infty$.

By adding these zeros we turned the integrals into closed contour integrals in the complex plane.

Now we can work out the two integrals over closed contours using the residue theorem. We pick up the residues of the poles of the integrands $\frac{e^{\pm iz}}{z}$ that are enclosed by the contour. The only pole is at $z = 0 + i\delta$. So the right integral picks up no residue, because C_2 does not enclose this pole. The left integral gives

$$\frac{1}{2i} \oint_{C_1} \frac{e^{iz}}{z} dz = 2\pi i \left(\lim_{\delta \to 0^+} \left. \frac{e^{iz}}{2i\pi} \right|_{z = i\delta} \right) = 1$$

$$\Rightarrow \quad \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin^2 t}{t^2} dt = 1 \quad \checkmark$$

Problem 3

- $\underbrace{\langle k|\hat{X}^2|f\rangle} = \int dx\, \langle k|\hat{X}^2|x\rangle\langle x|f\rangle = \int dx\, x^2 \langle k|x\rangle f(x)$

$$= \int \frac{dx}{\sqrt{2\pi}} e^{-ikx} x^2 f(x) = -\frac{d^2}{dk^2}\int\frac{dx}{\sqrt{2\pi}} e^{-ikx} f(x) = -\frac{d^2}{dk^2} f(k)$$

- $\langle x|\hat{X}^2|f\rangle \overset{\hat{X}=\hat{X}^+}{=} \langle \hat{X}^2 x|f\rangle = x^2\langle x|f\rangle = x^2 f(x)$

- $\langle k|\hat{P}^2|f\rangle \overset{\hat{K}=\hat{K}^+}{=} \hbar^2 \langle \hat{K}^2 k|f\rangle = \hbar^2 k^2 \langle k|f\rangle = \hbar^2 k^2 f(k)$

- $\langle x|\hat{P}^2|f\rangle = \hbar^2\int dk\, \langle x|\hat{K}^2|k\rangle\langle k|f\rangle = \hbar^2\int dk\, k^2\langle x|k\rangle f(k)$

$$= \hbar^2\int\frac{dk}{\sqrt{2\pi}} e^{ikx} k^2 f(k) = -\hbar^2\frac{d^2}{dx^2} f(x)$$

- $\langle x|\frac{1}{2}(\hat{X}\hat{P}+\hat{P}\hat{X})|f\rangle = \frac{\hbar}{2} x\langle x|\hat{K}|f\rangle + \frac{\hbar}{2}\int dx'\, \underbrace{\langle x|\hat{K}|x'\rangle}_{-i\delta(x-x')\frac{d}{dx'}}\underbrace{\langle x'|\hat{X}|f\rangle}_{x'f(x')}$

$$= \frac{\hbar}{2} x\int dx'\, \underbrace{\langle x|\hat{K}|x'\rangle}_{-i\delta(x-x')\frac{d}{dx'}}\underbrace{\langle x'|f\rangle}_{f(x')} - \frac{i\hbar}{2}\frac{d}{dx}(xf(x))$$

$$= -\frac{i\hbar}{2}\left[x\frac{df}{dx} + \frac{d}{dx}(xf(x))\right] = -i\hbar\left(x\frac{df}{dx} + \frac{1}{2}f(x)\right)$$

- $\langle k|\frac{1}{2}(\hat{X}\hat{P}+\hat{P}\hat{X})|f\rangle = \frac{\hbar}{2}\int dk'\, \underbrace{\langle k|\hat{X}|k'\rangle}_{i\delta(k-k')\frac{d}{dk'}}\underbrace{\langle k'|\hat{K}|f\rangle}_{k'f(k')} + \frac{\hbar}{2}k\langle k|\hat{X}|f\rangle$

$$= \frac{i\hbar}{2}\frac{d}{dk}(kf(k)) + \frac{\hbar}{2}k\int dk'\, \underbrace{\langle k|\hat{X}|k'\rangle}_{i\delta(k-k')\frac{d}{dk'}}\underbrace{\langle k'|f\rangle}_{f(k')}$$

$$= \frac{i\hbar}{2}\left(\frac{d}{dk}(kf(k)) + k\frac{df}{dk}\right) = i\hbar\left(k\frac{df}{dk} + \frac{1}{2}f(k)\right)$$